Class Activity # 3
Calculus Physics 2

me: Key Date: _____

Group Members: _____ _____ _____

Solve the following problems together as a group, stay engaged in the conversation with the group!

- Use **Lecture Notes Chapter 38 through 42** as a guide to answer the following questions.

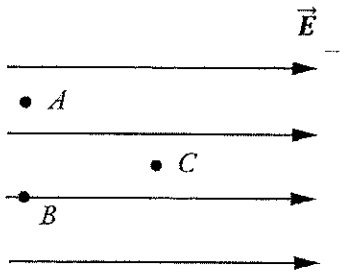
1) Suppose you have two point charges of opposite type. As you move them farther and farther apart, the potential energy of this system.



(A) increases.
B) decreases.
C) stays the same.

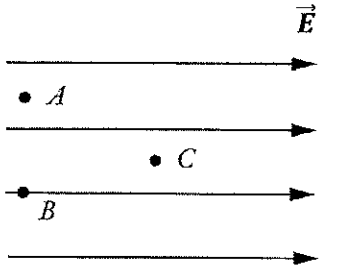
2) Suppose a region of space has a uniform electric field, directed towards the right, as shown in the figure. Which position will give an object with a net positive charge the most electric potential energy?

A) A
B) B
C) C
(D) Both A and B the same and larger than C.
E) The potential energy will be the same for the object at all points because it is a uniform electric field.



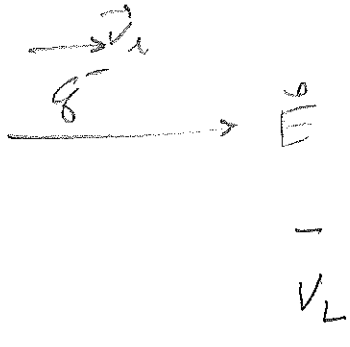
3) Suppose a region of space has a uniform electric field, directed towards the right, as shown in the figure. Which statement about the electric potential is true?

~~A)~~ The electric potential at all three locations (A, B, C) is the same because the field is uniform.
~~B)~~ The electric potential at points A and B are equal, and the potential at point C is higher than the potential at point A and B.
(C) The electric potential at points A and B are equal, and the potential at point C is lower than the potential at point A and B.
D) The electric potential at point A is the highest, the potential at point C is the second highest, and the potential at point B is the lowest.

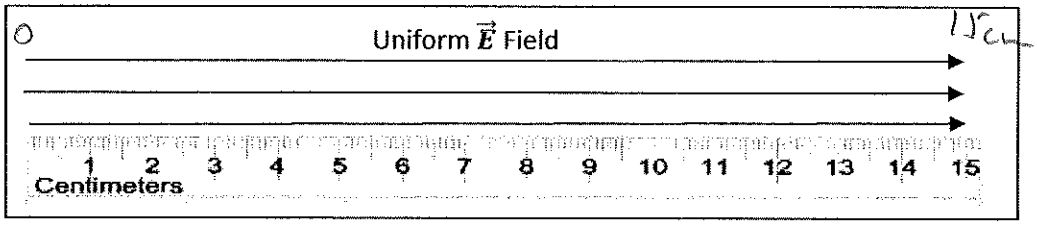


4) Which statements are true for an electron with an initial velocity moving in the direction of an electric field? (There is more than one correct choice.)

(A) Its electric potential energy increases as it goes from high to low potential.
~~B)~~ Its electric potential energy decreases as it goes from high to low potential.
(C) Its potential energy increases as its kinetic energy decreases.
(D) Its kinetic energy decreases as it moves in the direction of the electric field.
~~E)~~ Its kinetic energy increases as it moves in the direction of the electric field.



5) A ruler (not to scale) is placed along a uniform electric field with a magnitude of $|\vec{E}|$ = 800 N/C.



a) Given a sphere with a net positive charge of 500 μC, how much work does the Electric field do on the sphere to move the charged sphere from the 0 cm mark to the 15 cm mark? $|\vec{E}|$ = 800 N/C

$$W = \int_0^{\Delta x} \vec{F} \cdot d\vec{x} = q|\vec{E}|\Delta x = +0.06\,J$$

b) What was the change in Electric Potential energy of the sphere moving from the 0 cm mark to the 15 cm mark, and did it gain or loss PE?

$$\Delta PE = -W = -0.06\,J$$

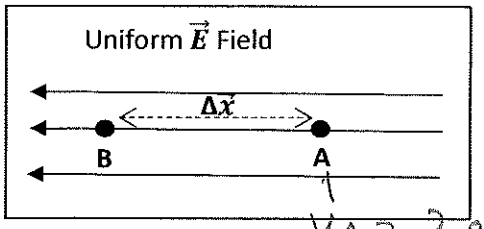
c) What Electric potential difference did the sphere move through when it moved from the 0 cm mark to the 15 cm mark?

$$\Delta V = \frac{\Delta PE}{q} = \frac{0.06\,J}{500\times10^{-6}\,C} = 120\,V$$

6) A sphere with a positive net charge of 12 μC is placed in a uniform electric field at point B and the distance between point A and Point B is 0.5 m.

A) The electric potential difference between Point A and Point B is 120 V, what is the magnitude of the electric field?

$$|\vec{E}| = \frac{\Delta V}{\Delta x} = 240\,V/m$$



B) If point A is at an electric potential of V_A = 200 V, what is the electric potential at Point B (V_B = ?)?

$$V_B = V_A - \Delta V_{AB} = 200V - 120V = 80V$$

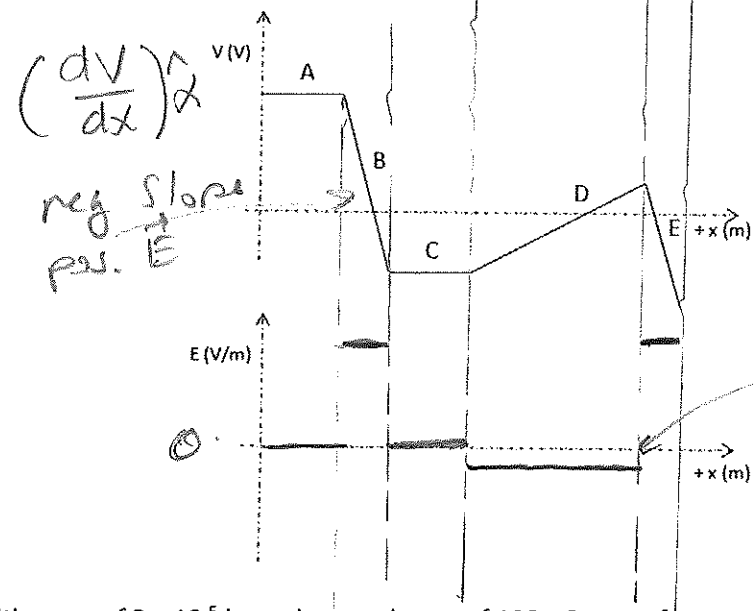
C) At what location would the electric potential be equal to zero and how far it the zero potential for the closes point (A or B)?

$$\Delta V = |\vec{E}|\Delta x_0 = V_B - 0$$

$$\Delta x_0 = \frac{V_B}{|\vec{E}|} = 0.33\,m \text{ to left of B}$$

7) Using the electric potential vs. position graph, draw the corresponding Electric field graph vs. position graph and label each of the sections indicated by the letters A- E.

$\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x}$



neg slope → pos. $\vec{E}$

positive slope → $-\vec{E}$ - smaller slope than B or E

8) A sphere with mass of 2×10^{-5} kg and a net charge of 100 μC starts from rest and is accelerated through a potential difference of 100 V, determine the final velocity of the sphere after exiting the 100 V?

$\Delta KE + \Delta PE = 0$

$\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$\Delta PE = -q\Delta V$

$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 - q\Delta V = 0$

$v_f = \sqrt{\frac{2q\Delta V}{m}} = 31.6\ m/s$

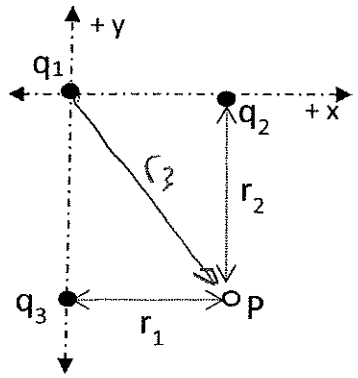
9) Three point charges are placed on the x and y axis as shown. Determine the Electric potential at point P due to the three charges relative to infinity?

$V_{Total} = \left(\frac{kq_1}{r_3}\right) + \left(\frac{kq_2}{r_2}\right) + \left(\frac{kq_3}{r_1}\right)$

$V_{Total} = (+228V) + (-60V) + (-180V)$

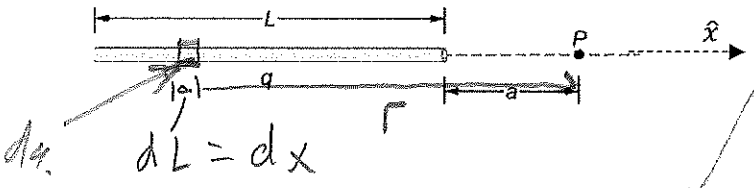
$V_{Total} = -12V$

$q_1 = 8$ nC positive
$q_2 = 2$ nC negative
$q_3 = 2$ nC negative
$r_1 = 0.1$ m
$r_2 = 0.3$ m
$r_3 = 0.316$ m



10) A thin wire of length L with a uniform linear charge density of λ is shown below, determine the electric potential at point P due to the charge on the wire. Draw and label on the diagram the incremental charge, radius, angle, and any other variables needed to solve the problem.

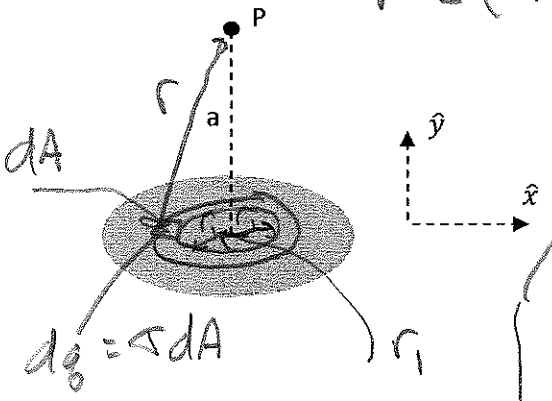
$r = x$ $\quad dq = \lambda dL = \lambda dx$



$V(r) = k\int \frac{dq}{r}$

$V(r) = k\int_a^{L+a} \frac{\lambda dx}{x} = k\lambda\int_a^{L+a}\left(\frac{1}{x}\right)dx = k\lambda\left[h(x)\right]_a^{L+a} = k\lambda\left[h(L+a) - h(a)\right]$

11) A circular disk with a radius R and a uniform surface charge density of σ is shown below, determine the electric potential at point P which is at a distance of "a" from the center of the disk. Draw and label on the diagram the incremental charge, radius, angle, and any other variables needed to solve the problem.



$r = (a^2 + r_1^2)^{1/2}$

$dA = (2\pi r_1)(dr_1)$

$dq = \sigma 2\pi r_1 dr_1$

$V(r) = K\int \frac{dq}{r} = K\int \frac{\sigma 2\pi r_1 dr_1}{(a^2+r_1^2)^{1/2}}$

$V(r) = K\sigma 2\pi \int_0^R \frac{r_1}{(a^2+r_1^2)^{1/2}} dr_1$

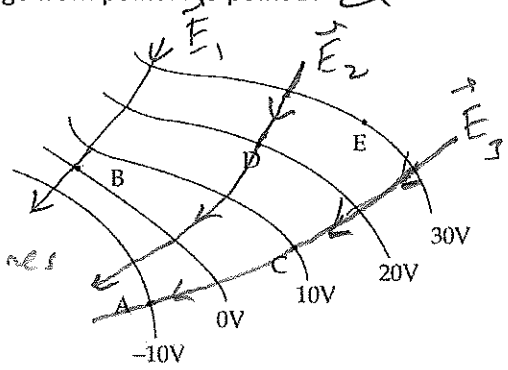
$V(r) = K\sigma 2\pi \left[\sqrt{r_1^2+a^2}\right]_0^R$

$V(r) = K\sigma 2\pi \left(\sqrt{R^2+a^2} - a\right)$

12) A negative charge is moved from point A to point B along an equipotential surface. Which of the following statements must be true for this case?

A) The negative charge performs work in moving from point A to point B.
B) Work is required to move the negative charge from point A to point B.
C) No work is required to move the negative charge from point A to point B.
D) The work done on the charge depends on the distance between A and B.
E) Positive work is done by the electric field in moving the negative charge from point A to point B.

13) Using the equipotential lines below answer the following questions.



A) What is the potential difference between points A and D?

$\Delta V_{AB} = 20V - (-10V) = 30V$

B) At what point is the electric field strength the greatest?

B → closest Spaced Equipotential lines

C) Draw 3 electric field lines and indicate it's direction on the diagram, field line # 1 should include point B, field line # 2 should include point D and field line # 3 should include point A.

14) Dielectric material such as nylon is inserted between the plates of a capacitor, the plates are NOT connected to a power supply. What happens to the electric field between the plates as the dielectric is inserted?

A) There is no change in the Electric field.
B) The Electric field becomes stronger.
C) The Electric field becomes weaker.
D) The Electric field reduces to zero.

0.4 mm

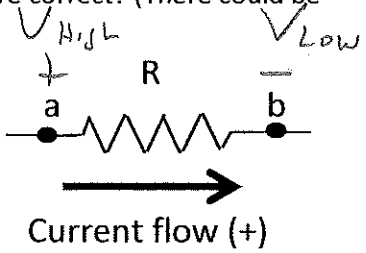
7 15) A parallel plate capacitor with plate separation of ~~4.0 cm~~ has a plate area of 0.06 m^2. Determine the capacitance of this capacitor if a dielectric material with a dielectric constant of 2.4 is placed between the plates?
$\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$ $\quad \varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$

$C = \kappa\left(\frac{\varepsilon_o A}{d}\right) = 3.2 \times 10^{-9}\ F$

16) An electric motor requires 2 J of electrical energy to start, if the capacitor is charged to 120 V, what must be the capacitance of the capacitor?

$Energy = \frac{1}{2} C V^2 \quad \Big| \quad C = \frac{2(Energy)}{V^2} = 2.8 \times 10^{-4}\ F$

17) The figure shows current passing through a resistor (R). The arrow shows the direction of the current (assume conventional current flow of positive charge). Which of the following statements are correct? (There could be more than one correct choice.)



A) There is more current at point b than at point *a*.
B) There is more current at point a than at point b.
C) The current is the same at point a and b.
D) The electric potential is higher at point b than at point a.
E) The electric potential is higher at point a than at point b.
F) The charge is losing electric potential energy as they move through the resistor from a to b.
G) The charge is gaining electric potential energy as they move through the resistor from a to b.

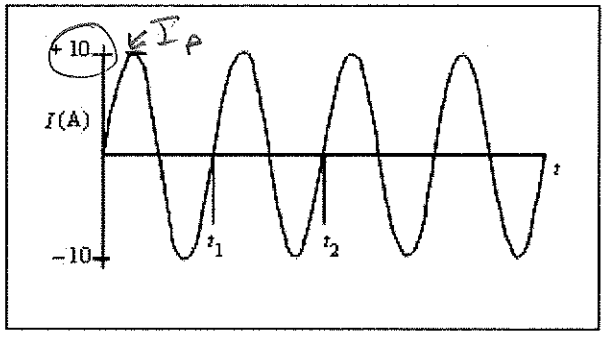
18) When electric current is flowing in a metal, the electrons are moving

A) at nearly the speed of light.
B) at the speed of light.
C) at the speed of sound in the metal.
D) at the speed of sound in air.
E) none of the above.

19) If the power used by the bulb is 5 W and the current through the bulb is 0.5 A, determine the resistance of the bulb?

$P = IV = I^2 R \quad \Big| \quad R = \frac{P}{I^2} = 20\ \Omega$

20) The figure shows variation of an a.c. current through a heating element over time in a toaster oven when it is plugged into a standard V_{rms} = 120 V at 60 Hz outlet.



A) What is the rms value of the current in this circuit?

$I_p = 10A \quad \Big| \quad I_{rms} = \frac{I_p}{\sqrt{2}} = 7.07A$

B) What is the resistance of the heating element?

$V_{rms} = I_{rms} R \quad \Big| \quad R = \frac{V_{rms}}{I_{rms}} = 17\ \Omega$

C) What is the average power dissipated by the heating element?

$P_{av} = I_{rms} V_{rms} = 848\ W$

21) A cylindrical wire has a resistance R and resistivity ρ. If its length and ~~diameter~~ radius are BOTH cut in half,

(a) What will be its resistance?

A) $4R$
B) $2R$
C) R
D) $R/2$
E) $R/4$

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$$

$$R_1 = \frac{\rho(\frac{1}{2}L)}{\pi(\frac{1}{2}r)^2} = (\frac{1}{2})(4)\frac{\rho L}{\pi r^2} = 2R$$

(b) What will be its resistivity?

A) 4ρ
B) 2ρ
C) ρ
D) $\rho/2$
E) $\rho/4$

ρ → material dependent
not Length or radius

22) For most metallic conductors, increasing the temperature causes:

A) Resistance to decrease because electrons move faster.
B) Resistance to remain constant.
C) Resistance to increase.
D) Resistance becomes zero.

23) 4 meters of Ni-chrome cylinder wire ($\rho = 100 \times 10^{-8}\ \Omega$) with a radius of 2×10^{-4} m is wired to a 120 V power source. The temperature coefficient of Ni-chrome is $\alpha = 4 \times 10^{-4}$ /°C.

A) Determine the power dissipated by the wire at 20 °C (room temperature)

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2} = 31.8\ \Omega$$

$$V = IR \rightarrow I = \frac{V}{R} = 3.77\ A$$

$$P = IV$$

$$P = 452.4\ W$$

B) Determine the power dissipated by the wire at 2000 °C (glowing yellow).

$$R_{hot} = R_0[1 + \alpha(T - T_0)] = (31.8\ \Omega)(1 + 2(2000°C - 20°C))$$

$$R_{hot} = 57\ \Omega$$

$$V = IR \rightarrow I = \frac{V}{R} = 2.1\ A$$

$$P = IV = 252.6\ W$$ ← Lower Power at higher temp!